3 Subject content

This syllabus gives you the flexibility to design a course that will interest, challenge and engage your learners. Where appropriate you are responsible for selecting resources and examples to support your learners' study. These should be appropriate for the learners' age, cultural background and learning context as well as complying with your school policies and local legal requirements.

Computer Science is a practical subject and a range of practical exercises must be integral to the teaching of this qualification. It is important that learners develop their computational thinking skills by doing practical problem-solving and programming using appropriate resources. It is also expected that learners have the opportunity in class to write their own programs, as well as executing (running), testing and debugging them.

Any equipment and facilities should be adequate for learners to be able to satisfy the requirements of the syllabus. The hardware facilities needed will depend on the number of learners but must be sufficient for all learners to have enough time to practise their programming skills. Learners also need to have access to a system with direct-access file capability on backing store and hardcopy facilities.

Computer systems

1 Data representation

1.1 Number systems

Candidates should be able to:	Notes and guidance
1 Understand how and why computers use binary to represent all forms of data	• Any form of data needs to be converted to binary to be processed by a computer • Data is processed using logic gates and stored in registers
2 (a) Understand the denary, binary and hexadecimal number systems	• Denary is a base 10 system • Binary is a base 2 system • Hexadecimal is a base 16 system
(b) Convert between (i) positive denary and positive binary (ii) positive denary and positive hexadecimal (iii) positive hexadecimal and positive binary	• Values used will be integers only • Conversions in both directions, e.g. denary to binary or binary to denary • Maximum binary number length of 16-bit
3 Understand how and why hexadecimal is used as a beneficial method of data representation	• Areas within computer science that hexadecimal is used should be identified • Hexadecimal is easier for humans to understand than binary, as it is a shorter representation of binary
4 (a) Add two positive 8-bit binary integers (b) Understand the concept of overflow and why it occurs in binary addition	• An overflow error will occur if the value is greater than 255 in an 8-bit register • A computer or a device has a predefined limit that it can represent or store, for example 16-bit • An overflow error occurs when a value outside this limit should be returned

1.1 Number systems continued

Candidates should be able to:	Notes and guidance
5 Perform a logical binary shift on a positive 8-bit binary integer and understand the effect this has on the positive binary integer	• Perform logical left shifts • Perform logical right shifts • Perform multiple shifts • Bits shifted from the end of the register are lost and zeros are shifted in at the opposite end of the register • The positive binary integer is multiplied or divided according to the shift performed • The most significant bit(s) or least significant bit(s) are lost
6 Use the two's complement number system to represent positive and negative 8-bit binary integers	• Convert a positive binary or denary integer to a two's complement 8-bit integer and vice versa • Convert a negative binary or denary integer to a two's complement 8-bit integer and vice versa

1.2 Text, sound and images

Candidates should be able to:	Notes and guidance
1 Understand how and why a computer represents text and the use of character sets, including American standard code for information interchange (ASCII) and Unicode	• Text is converted to binary to be processed by a computer • Unicode allows for a greater range of characters and symbols than ASCII, including different languages and emojis • Unicode requires more bits per character than ASCII
2 Understand how and why a computer represents sound, including the effects of the sample rate and sample resolution	• A sound wave is sampled for sound to be converted to binary, which is processed by a computer • The sample rate is the number of samples taken in a second • The sample resolution is the number of bits per sample • The accuracy of the recording and the file size increases as the sample rate and resolution increase

1.2 Text, sound and images continued

Candidates should be able to:	Notes and guidance
3 Understand how and why a computer represents an image, including the effects of the resolution and colour depth	• An image is a series of pixels that are converted to binary, which is processed by a computer • The resolution is the number of pixels in the image • The colour depth is the number of bits used to represent each colour • The file size and quality of the image increase as the resolution and colour depth increase

1.3 Data storage and compression

Candidates should be able to:	Notes and guidance
1 Understand how data storage is measured	• Including: – bit – nibble – byte – kibibyte (KiB) – mebibyte (MiB) – gibibyte (GiB) – tebibyte (TiB) – pebibyte (PiB) – exbibyte (EiB) • The amount of the previous denomination present in the data storage size, e.g.: – 8 bits in a byte – 1024 mebibytes in a gibibyte
2 Calculate the file size of an image file and a sound file, using information given	• Answers must be given in the units specified in the question. Calculations must use the measurement of 1024 and not 1000 • Information given may include: – image resolution and colour depth – sound sample rate, resolution and length of track
3 Understand the purpose of and need for data compression	• Compression exists to reduce the size of the file • What the impact of this is, e.g.: – less bandwidth required – less storage space required – shorter transmission time

1.3 Data storage and compression continued

Candidates should be able to:	Notes and guidance
4 Understand how files are compressed using lossy and lossless compression methods	• Lossy compression reduces the file size by permanently removing data, e.g. reducing resolution or colour depth, reducing sample rate or resolution • Lossless compression reduces the file size without permanent loss of data, e.g. run length encoding (RLE)

2 Data transmission

2.1 Types and methods of data transmission

Candidates should be able to:	Notes and guidance
1 (a) Understand that data is broken down into packets to be transmitted	
(b) Describe the structure of a packet	• A packet of data contains a: – packet header – payload – trailer • The packet header includes the: – destination address – packet number – originator's address
(c) Describe the process of packet switching	• Data is broken down into packets • Each packet could take a different route • A router controls the route a packet takes • Packets may arrive out of order • Once the last packet has arrived, packets are reordered
2 (a) Describe how data is transmitted from one device to another using different methods of data transmission	• Including: – serial – parallel – simplex – half-duplex – full-duplex
(b) Explain the suitability of each method of data transmission, for a given scenario	• Including the advantages and disadvantages of each method
3 Understand the universal serial bus (USB) interface and explain how it is used to transmit data	• Including the benefits and drawbacks of the interface